ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-1 **SUMMER SEMESTER, 2023-2024**

DURATION: 35 MINUTES **FULL MARKS: 15**

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. Suppose, R and k are arbitrary numbers such that $R > 0$ and $k \neq 0$. Prove that the Newton-Raphson formula for finding the inverse of the kth root of R, which is mathematically denoted as $\frac{1}{\sqrt[k]{R}}$, is as shown in Equation 1. 3 (CO2) (PO1)

$$x_{i+1} = \frac{1}{k}\left[(k-1)x_i + \frac{1}{Rx_i^{k-1}}\right] \tag{1}$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} x &= \frac{1}{\sqrt[k]{R}} \\ \Rightarrow \quad x^k &= \frac{1}{R} \\ \Rightarrow \quad x^k - \frac{1}{R} &= 0 \end{aligned}$$

We take $f(x) = x^k - \frac{1}{R}$ and so $f'(x) = kx^{k-1}$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^k - \frac{1}{R}}{kx_i^{k-1}} \\ &= x_i - \frac{Rx_i^k - 1}{Rkx_i^{k-1}} \\ &= x_i - \left(\frac{Rx_i^k}{Rkx_i^{k-1}} - \frac{1}{Rkx_i^{k-1}}\right) \\ &= x_i - \left(\frac{x_i}{k} - \frac{1}{Rkx_i^{k-1}}\right) \\ &= \frac{kx_i - x_i}{k} + \frac{1}{Rkx_i^{k-1}} \\ &= \frac{(k-1)x_i}{k} + \frac{1}{Rkx_i^{k-1}} \\ &= \frac{1}{k}\left[\frac{(k-1)x_i}{k} + \frac{1}{Rx_i^{k-1}}\right] \end{aligned}$$

□ Q.E.D.

Rubric:

- 1 point for using the Newton-Raphson approximation formula.
- 2 points for correct derivation.

2. a) Derive the formula to calculate m as a function of $|\epsilon_a|\%$ and determine the minimum threshold for the absolute relative approximate error ($|\epsilon_a|\%$) that would result in a negative value for m (which practically indicates that not even one significant digit is reliable). 2 (CO1) (PO2)

Solution:

To guarantee that m significant digits are correct, the absolute relative approximate error must be less than or equal to the pre-specified tolerance (ϵ_s):

$$|\epsilon_a|\% \leq 0.5 \times 10^{2-m}\%$$

To isolate m, first multiply both sides by 2:

$$2 \times |\epsilon_a| \leq 10^{2-m}$$

Next, take the base-10 logarithm of both sides:

$$\log_{10}(2 \times |\epsilon_a|) \leq \log_{10}\left(10^{2-m}\right)$$
$$\log_{10}(2 \times |\epsilon_a|) \leq 2 - m$$

Rearranging the inequality to solve for m yields:

$$m \leq 2 - \log_{10}(2 \times |\epsilon_a|)$$

Because m represents a discrete number of digits, it must be the largest integer that satisfies this condition. Therefore, we apply the floor function:

$$m = \lfloor 2 - \log_{10}(2 \times |\epsilon_a|) \rfloor$$

For m to be negative ($m < 0$), the expression inside the floor function from our derivation must be strictly less than 0. Setting up the inequality:

$$2 - \log_{10}(2 \times |\epsilon_a|) < 0$$

Solving for $|\epsilon_a|$:

$$2 < \log_{10}(2 \times |\epsilon_a|)$$

Converting from logarithmic to exponential form:

$$10^2 < 2 \times |\epsilon_a|$$
$$100 < 2 \times |\epsilon_a|$$
$$|\epsilon_a| > 50$$

Therefore, m will become negative for any absolute relative approximate error strictly greater than 50%.

Rubric:

- 1 point for the correct formula.
- 1 *bonus* point for the correct minimum threshold.

b) Determine the *minimum* number of significant digits in each of these numerical values — 3 (CO4) (PO1)

(a) 69.0 (b) 69,420 (c) 6.94200×10^{911}

(d) 69.00420 (e) 069420 (f) 0.069420

Solution:

We need to determine the *minimum* number of significant digits.

(i) 69.0 has 3 significant digits, since 0's to the right of the decimal point are significant.

(ii) $69{,}420$ has 4 significant digits, since trailing 0's in an integer may or may not be significant.

(iii) 6.94200×10^{911} has 6 significant digits, since all digits of the significand/coefficient of a number's representation using Scientific Notation are significant.

(iv) 69.00420 has 7 significant digits, since 0's between two significant non-zero digits are significant and trailing 0's in a number with the decimal point are significant.

(v) 069420 has 4 significant digits, since leading 0's are not significant and trailing 0's in an integer may or may not be significant.

(vi) 0.069420 has 5 significant digits, since leading 0's are not significant and trailing 0's in a number with the decimal point are significant.

Rubric:

- 0.5 point for each correct answer. (The reasons are not required.)

3. "If a function $f(x)$ does **not** change its sign between two points x_l and x_u, then there may or may not be any root of the equation $f(x) = 0$ between them." — Do you agree? Justify your answer. 3 (CO3) (PO2)

Solution:

Yes, I agree.

There are 4 possible cases for which we get $f(x_l)f(x_u) > 0$. If we examine the 1st and 2nd cases (as shown in Figure-1 and Figure-2 respectively), we observe that there can be multiple roots between x_l and x_u.

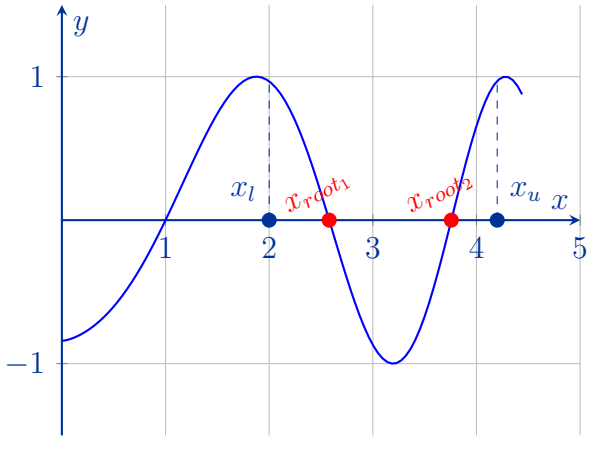


Figure 1: The 1st case (both $f(x_l)$ and $f(x_u)$ are positive, and roots exist).

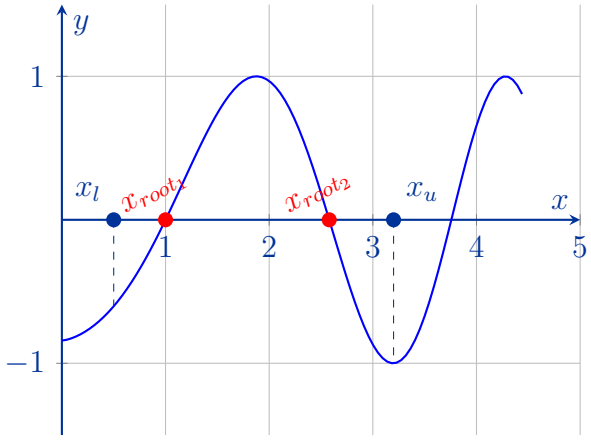


Figure 2: The 2nd case (both $f(x_l)$ and $f(x_u)$ are negative, and roots exist).

If we examine the 3rd and 4th cases (as shown in Figure-3 and Figure-4 respectively), we observe that there are no roots between x_l and x_u.

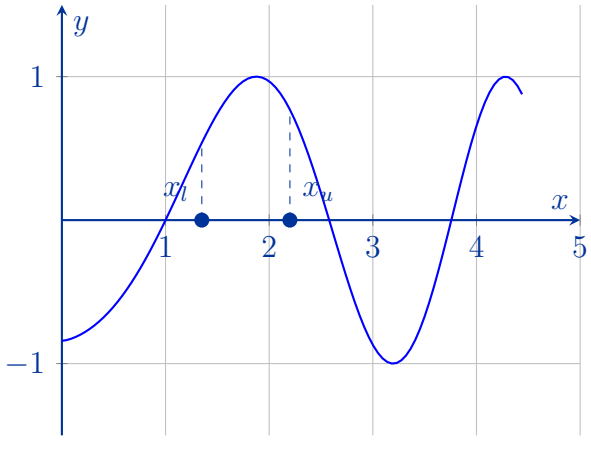


Figure 3: The 3rd case (both $f(x_l)$ and $f(x_u)$ are positive, but roots don't exist).

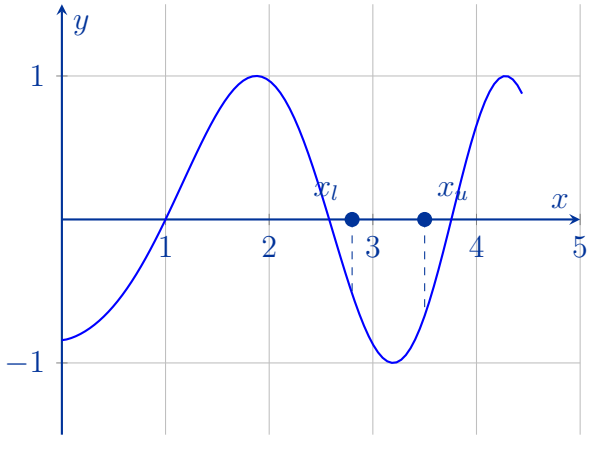


Figure 4: The 4th case (both $f(x_l)$ and $f(x_u)$ are negative, but roots don't exist).

Rubric:

- 1 point for the correct verdict.
- 0.5 point for each possible case.

4. As depicted in Figure 5, which illustrates the velocity components acting on a helicopter rotor disk in forward flight, the inflow ratio λ (dimensionless) can be computed using Equation 2.

$$\lambda = \mu \tan(\alpha_c) + \frac{C_T}{2\sqrt{\mu^2 + \lambda^2}} \tag{2}$$

where C_T is the rotor thrust coefficient, μ is the advance ratio (dimensionless forward speed), and α_c is the rotor disk angle of attack (rad). For a helicopter in a steady climb, the rotor disk angle of attack is non-zero, making the calculation more complex.

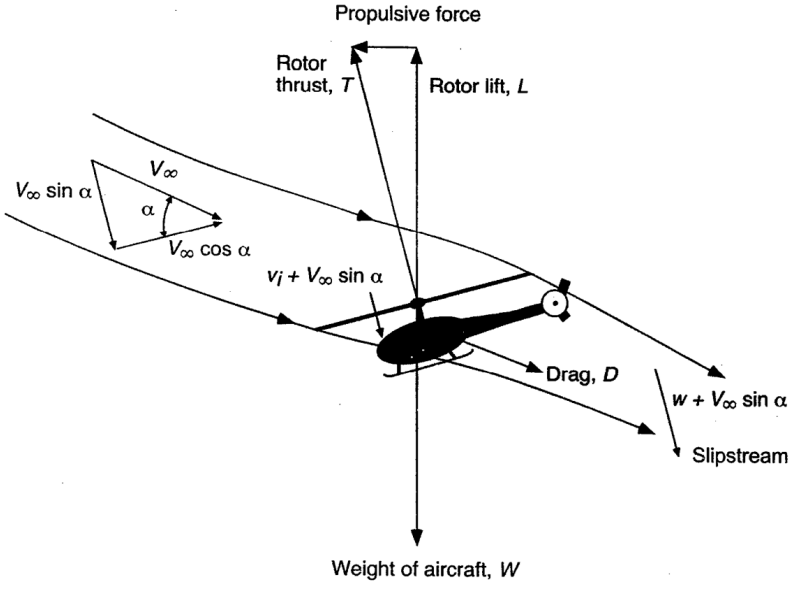


Figure 5: Velocity components and angles for a helicopter rotor disk in forward flight en route to IUT.

a) Formulate a function $f(\lambda)$ such that its root will yield the value of the inflow ratio λ for a climbing flight condition, given values of C_T, μ, and α_c. 1 (CO1) (PO2)

Solution:

The given equation is,

$$\lambda = \mu \tan(\alpha_c) + \frac{C_T}{2\sqrt{\mu^2 + \lambda^2}}$$

By moving all terms to one side of the equation, we get:

$$\lambda - \mu \tan(\alpha_c) - \frac{C_T}{2\sqrt{\mu^2 + \lambda^2}} = 0$$

So, the function $f(\lambda)$ such that its root will yield the value of the inflow ratio λ for given values of C_T, μ, and α_c is as follows:

$$f(\lambda) = \lambda - \mu \tan(\alpha_c) - \frac{C_T}{2\sqrt{\mu^2 + \lambda^2}}$$

Rubric:

- 1 point for the correct function.

b) Apply the Secant method to estimate the inflow ratio λ that is needed to maintain the climb for a helicopter with a thrust coefficient $C_T = 0.008$, an advance ratio $\mu = 0.20$, and a rotor disk angle of attack $\alpha_c = 0.05\,\text{rad}$. The initial guesses are $\lambda_{-1} = 0.20$ and $\lambda_0 = 0.30$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at 4 (CO2) (PO1)

the end of each iteration. Draw Table 1 on your answer script and fill it out after performing the necessary calculations.

Table 1: The relevant values obtained in the answer of Question 4b

Iteration	λ_{i-1}	λ_i	λ_{i+1}	$\lvert\epsilon_a\rvert\%$	m	$f(\lambda_{i+1})$
1	0.20	0.30				
2						
3						
4						

Solution:

Substituting $C_T = 0.008$, $\mu = 0.20$, and $\alpha_c = 0.05$ into the formulated function:

$$f(\lambda) = \lambda - 0.20\tan(0.05) - \frac{0.008}{2\sqrt{0.04+\lambda^2}}$$

$$f(\lambda) \approx \lambda - 0.01001 - \frac{0.004}{\sqrt{0.04+\lambda^2}}$$

After performing the series of calculations using the Secant method formula $\lambda_{i+1} = \lambda_i - f(\lambda_i)\frac{\lambda_i - \lambda_{i-1}}{f(\lambda_i) - f(\lambda_{i-1})}$, we end up with the following table.

Iteration	λ_{i-1}	λ_i	λ_{i+1}	$\lvert\epsilon_a\rvert\%$	m	$f(\lambda_{i+1})$
1	0.20000	0.30000	0.02936	921.798	0	-0.00044
2	0.30000	0.02936	0.02893	1.486	1	-0.00087
3	0.02936	0.02893	0.02980	2.919	1	0.00001
4	0.02893	0.02980	0.02979	0.034	3	0.00000

Rubric:

- 0.25 point for each *approximately* correct answer.